

BILAL ZAHID

AI Infrastructure Engineer | Agentic Systems & LLM Evaluation

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Professional Summary

AI infrastructure engineer and technical lead with 6+ years building production systems end-to-end. Architected Google's Gemini training and evaluation infrastructure at Turing, agentic simulation over 77 real-world APIs, synthetic-world generation, and RLHF/SFT across Gemini, Grok, and ServiceNow AI. Ships distributed, cloud-native platforms across enterprise SaaS and healthcare while leading engineering teams and shaping technical direction.

Professional Experience

Google Services - LLM Python Developer



06/2024 - Present

Core engineer on Google's Gemini training and evaluation infrastructure, the internal "Gemini Gym", with parallel work on LLM fine-tuning and evaluation for xAI's Grok and ServiceNow AI.

- **Architected the Gym's agentic simulation environment (Generalized Agent APIs):** rebuilt 77 real-world APIs as a deterministic, resettable sandbox with 3,200+ automated tests, so Gemini trains and is scored on multi-turn tool calling without touching live data.
- **Designed a multi-stage function-calling validation framework** across 75+ tool schemas, enforcing schema integrity, cross-provider compatibility (Google, OpenAI, Anthropic), and function-signature consistency, catching defects Google's own GenAI SDK missed and raising automated quality coverage to roughly 90%.
- **Built a sim-to-real fidelity and regression suite** across 36+ services, proving simulated tools matched production API behavior, cutting downstream defects 20%, and wiring validation directly into CI.
- **Engineered DBGen, the Gym's synthetic-world generation pipeline:** transforms a one-paragraph specification into a fully-populated, schema-accurate training world with an auto-generated evaluation rubric, validated across three generations of Gemini models.
- **Designed the Gym's robustness-evaluation harness (Benchmark Mutation Suite):** degrades tool schemas, input modality, and call format across 10 industry benchmarks, including the 109-task Toolathlon and 231-task MCP Universe suites, benchmarking Gemini head-to-head against GPT-5 and Claude under real-world failure conditions.
- **Led supervised fine-tuning (SFT) and RLHF data curation** for Gemini, Claude, Grok, and ServiceNow AI, producing code-centric training datasets that raised contextual accuracy 18% and cut inference latency 45 ms.
- **Led a 6-person team of AI trainers and engineers** across dataset curation, infrastructure, and model evaluation, setting sprint priorities and reviewing code to keep delivery on schedule.

Stack: Python · React · Next.js · Django · FastAPI · Pydantic · TensorFlow · PyTorch · Hugging Face · AWS · Docker · SQL · Pinecone · Prompt Engineering · RLHF/SFT.

Senior FullStack Engineer



09/2022 - 09/2024

Delivered across three enterprise SaaS products in a single tenure, EdTech, AI-powered revenue operations, and global indirect-tax compliance, each engineered cloud-native with intelligent workflow automation.

- **AI-Powered Revenue Operations (GTM):** built autonomous multi-agent CRM workflows on a serverless, event-driven AWS Lambda architecture with NLP-driven pipelines and React/Next.js analytics dashboards, lifting sales-team productivity 30%, operational efficiency 250%, and close-rate 150%.
- **Global Tax Compliance Platform:** founding engineer on the platform, architecting registration, calculation, and filing automation with geolocation-based jurisdiction detection across international jurisdictions.
- **EdTech Learning Platform:** led an earlier live learning platform (Ruby on Rails, React/Next.js) with real-time class tools and a companion Zoom note-taking app, lifting user engagement 40%.
- **Team leadership (across tenure):** owned system reliability and engineering quality for a 7-person team, tuning PostgreSQL, Pinecone, and MongoDB (bugs down 20%, maintainability up 15%), drove technical roadmap discussions with the CEO and CTO, and mentored 5 engineers through onboarding and delivery.

Stack: Ruby on Rails · Python (FastAPI) · React · Next.js · TypeScript · OpenAI · AWS (Lambda, RDS, EC2, S3, CloudWatch, API Gateway, SES, CDN) · Salesforce · HubSpot · PostgreSQL · MongoDB · Pinecone · Redis · Docker · Stripe · Fivetran · PropelAuth.

Onboarding: completed a three-month internal tooling project (MapleHR, an internal HR/payroll & inventory system) before client placement.

Senior Software Engineer

 Duett.io

01/2021 - 08/2022

Built a HIPAA-compliant healthcare marketplace connecting care coordinators with home-health providers to streamline patient care across Central Indiana.

- **Architected a HIPAA-compliant, cloud-native healthcare platform (DUETT)** solving a core care-coordination and interoperability problem, connecting care coordinators with home-health providers through a resilient backend and reshaping care delivery across Central Indiana.
- **Designed a provider-matching engine** on Django and Flask that translated patient needs into optimized care-coordinator assignments, backed by rigorous safeguards for protected health information.
- **Shipped scheduling and workflow-optimization features** that lifted sales 15% and cut bounce rate 25% on a React and Django stack.
- **Integrated predictive-analytics models**, developed with data science partners, into the provider-matching pipeline, raising match accuracy 20%.
- **Scaled backend services on AWS** to sustain 10,000 requests per minute while cutting API latency 25%, keeping the platform resilient through provider-network growth.
- **Expanded Duett's provider network** by integrating case managers and clients into the platform, establishing Duett as a recognized leader in healthcare-service matching.

Stack: Python (Django, Flask) · React · PostgreSQL · MongoDB · AWS (EC2, S3, SES) · Jenkins · Grafana · MUI.

Software Engineer

 JoinReflect

03/2020 - 01/2021

Digital mental-health platform that matches patients to the right therapists and coordinates their care.

- **Built a digital health platform ("Reflect")** powered by an intelligent recommendation algorithm that matched patients to therapists based on specific stressors, lifting match accuracy 30%.
- **Designed a calendar-orchestration microservice** on Django handling asynchronous scheduling for therapist availability, eliminating scheduling conflicts 40% and shipping as a standalone, reusable service.
- **Engineered secure, scalable microservices** and RESTful APIs on AWS within a cross-functional Agile team, sustaining 10,000 requests per minute while cutting data-retrieval time 25%.

Stack: Python (Django) · React · AWS (EC2, Lambda, RDS, CloudWatch, CDN) · MongoDB · PostgreSQL · MUI · Mailgun.

Technical Skills

AI / LLM: Agentic AI, Tool Calling, Function Calling, Function Schema Validation, Synthetic Data Generation, LLM Evaluation, Evaluation Pipelines, Simulation Environments, Benchmark Harnesses, World Generation, Multi-Turn Reasoning, RLHF, SFT, Prompt Engineering, Model Evaluation, Robustness Testing, LLM Orchestration, Hugging Face, PyTorch, TensorFlow

Languages: Python, TypeScript, JavaScript, SQL, Ruby, GraphQL, HTML5, CSS3

Backend: FastAPI, Django, Flask, Ruby on Rails, REST APIs, GraphQL, Pydantic, Protocol Buffers, NLP, Microservices, Event-Driven Architecture

Cloud & DevOps: AWS (EC2, Lambda, ECS, S3, RDS, CloudWatch, API Gateway, SES, CDN), GCP, Cloud Architecture, Docker, Kubernetes, CI/CD, CircleCI, Jenkins, Heroku, Datadog, Grafana, Observability, Linux

Databases: PostgreSQL, MongoDB, MySQL, Pinecone, Redis, Elasticsearch

Frontend: React, Next.js, TypeScript, jQuery, Material UI (MUI), Bootstrap

Practices & Tools: System Design, Distributed Systems, Agile/Scrum, HIPAA Compliance, OWASP, API Design, Performance Optimization, Apache JMeter, Jest, Amplitude, Figma, Git/GitHub

Education

FAST-NUCES (National University of Computer & Emerging Sciences)

Lahore, Pakistan

Bachelor of Science in Computer Science

08/2016 - 05/2020